

Digital Integrated Circuits

Homework#6 2024.12.12 (Due: 2025.01.08 23:59 ADFP Cloud 2.0)

- **Using ADFP standard V_T technology with $V_{DD} = 0.8V$**
- **Tutorial File:**
/RAID2/COURSE/2024_Fall/DIC/DICTA01/DIC_Final_Submission/DIC2024_ADFP-Layout-Tutorial.pdf
- **Design Rules:** [~/ADFP_PDF/ADFP039_N16ADFP_DRM_V1.1_1.pdf](#)

- (1) Design an inverter with arbitrary number of fin in Virtuoso using:
 - i. Schematic with Pre-Sim Simulation. (Input Waveform $0 \rightarrow 1 \rightarrow 0$ with rise time $0V \sim 0.8V$ and fall time $0.8V \sim 0V$ equal to 10ps.) (10%)
 - ii. Layout with 0 DRC error (except dummy and density issues), 0 LVS error, and Post-Sim Simulation. (20%)
- (2) Design a 4-bit **Ripple Carry Adder** in Virtuoso using:
 - i. Schematic with Pre-Sim Simulation. (rise time and fall time = 10ps with following input signals) (20%)

Pattern	time	A[3:0]	B[3:0]	C_{in}	Out[4:0]
0	0ps	4'b0000	4'b0000	1'b0	5'b00000
1	10ps	4'b1010	4'b0101	1'b1	5'b10000
2	10ps + 2T	4'b1100	4'b0011	1'b0	5'b01111
3	10ps + 3T	4'b1111	4'b1111	1'b1	5'b11111

(Note: T = maximum propagation delay of your adder)

- ii. Layout with 0 DRC error (except dummy and density issues), 0 LVS error, and Post-Sim Simulation. (30%)
- (3) Performance Ranking (Area * Propagation Delay) (20%)
 - i. Area of Chip Boundary (μm^2)
 - ii. Propagation Delay (ps): (Delay of **[Pattern₁]** + **[Pattern₂]** + **[Pattern₃]**)

Submission:

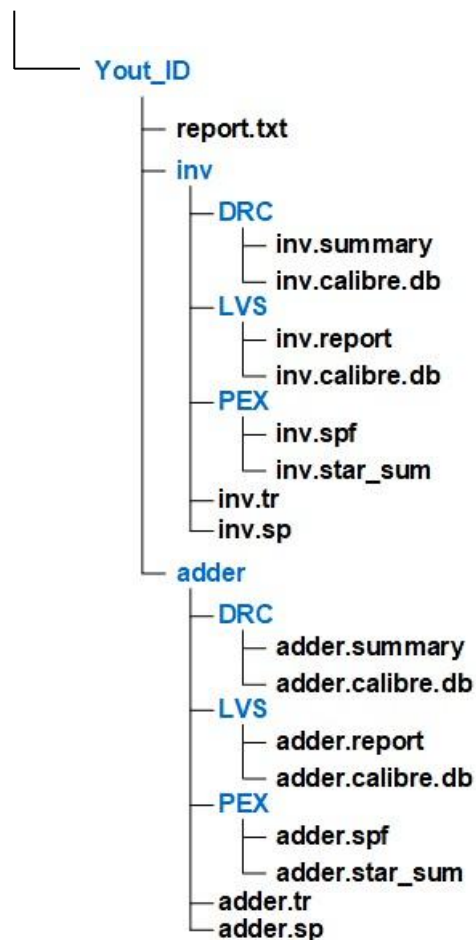
1. Make a directory using **Your_ID**
2. Create **inv** directory, **adder** directory, and copy **report.txt** in **Your_ID**
Template file of **report.txt**:
/RAID2/COURSE/2024_Fall/DIC/DICTA01/DIC_Final_Submission/report.txt
3. create **DRC**, **LVS**, **PEX** directories and organize them as the hierarchy below.

4. Edit **report.txt** and make sure you have submitted all the files.
5. Link submission scripts: `ln -s ~DICTA03/01_submit`
`ln -s ~DICTA03/02_check`
6. Create a “**HW6_submit**” folder: `mkdir HW6_submit`
“HW6_submit” should be created in the same directory as the scripts.
7. Put your submission files in “**HW6_submit**”.

01_submit

02_check

HW6_submit



8. Run 01_submit: `./01_submit`

```

[Info] Now start tar zcvf HW6_submit_course.tar.gz
HW6_submit/
HW6_submit/Tutorial.jpg
[Success] HW6_submit_course.tar.gz
[Info] Deadline check OK ...
[Info] File check OK ...
[Info] Your file will be submitted to: HW6
[Info] Are you sure you want to submit your design file?(y/n):
  
```

9. Type “y” to continue.

Please make sure there are no errors, such as “file not found”, after running ./01_submit.

```
[Info] Now start tar zcvf HW6_submit_course.tar.gz
HW6_submit/
HW6_submit/Tutorial.jpg
[Success] HW6_submit_course.tar.gz
[Info] Deadline check OK ...
[Info] File check OK ...
[Info] Your file will be submitted to: HW6
[Info] Are you sure you want to submit your design file?(y/n):y
[Info] Now submit HW6_submit_course.tar.gz file to system.
[Success] Copying Sucessfully.

=====
Submit Report
=====
Result      : HW6 has been submitted.
Submission time : Mon Dec 16 04:31:28 PM CST 2024
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--              --      | \_||
-- Congratulations !! -- / 0.0 |
--              --      | /---|
-- Submission Sucessful!!-- / ^ ^ ^ \ |
--              --      | ^ ^ ^ ^ |w|
--              --      | \m__m__|_|
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Please remember to check your submission with ./02_check !!
Please remember to check your submission with ./02_check !!
Please remember to check your submission with ./02_check !!
=====
course@icssl-4090:~/Project/312510232/File_Submission$
```

Make sure there are no any error

10. Run 02_check: ./02_check

The submission file will be downloaded in the “HW6_submit_[your server account]_check” folder.

Please make sure there are no errors, such as “file not found”, after running ./02_check.

```
HW6_submit_course.tar.gz has been downloaded!
It was put in the HW6_submit_course_check folder!!
```